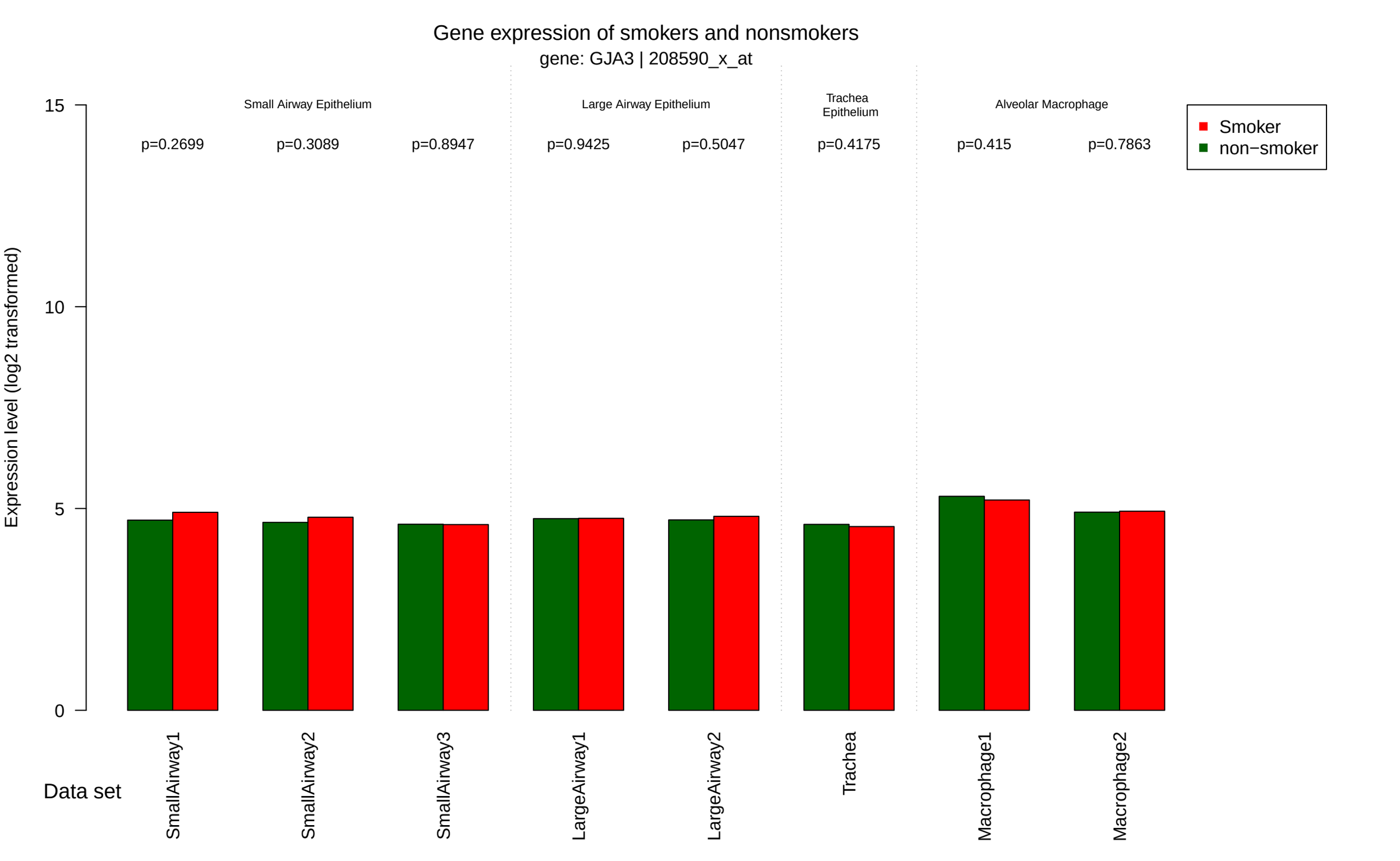
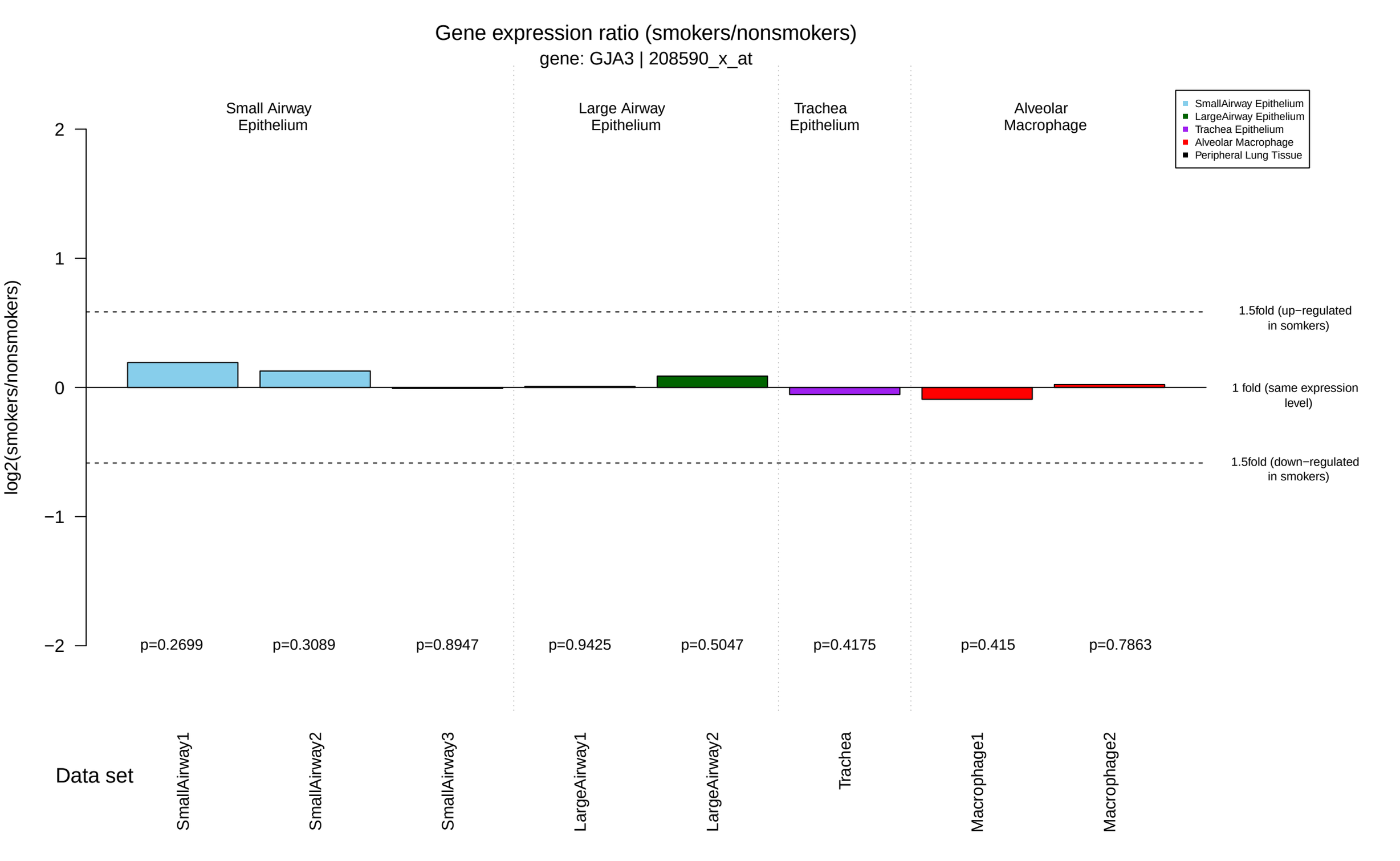
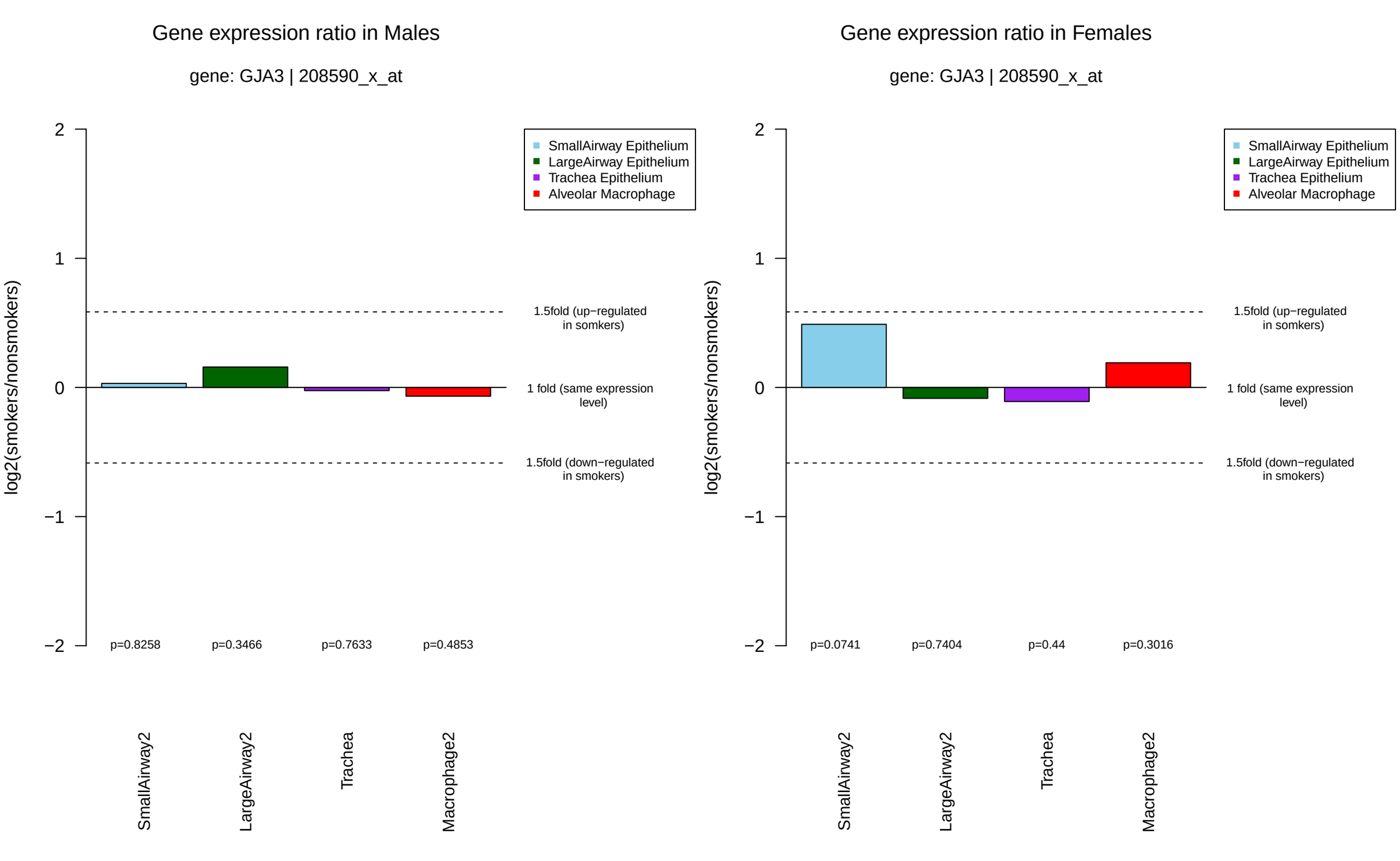
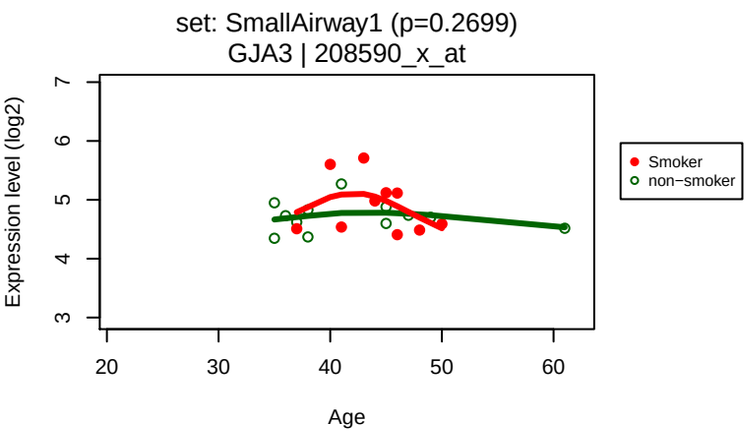
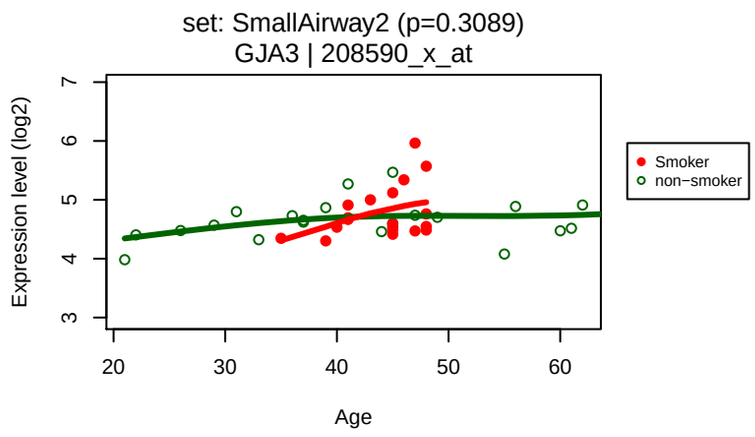
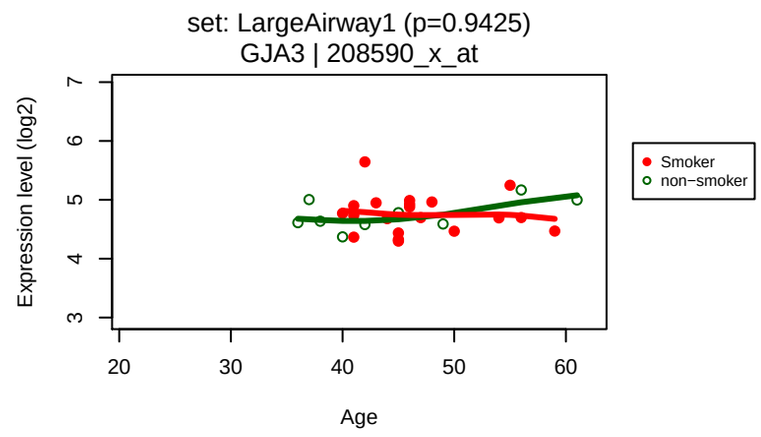

Gene expression of smokers and nonsmokers
gene: GJA3 | 208590_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2699
p=0.3089
p=0.8947
p=0.9425
p=0.5047
p=0.4175
p=0.415
p=0.7863
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

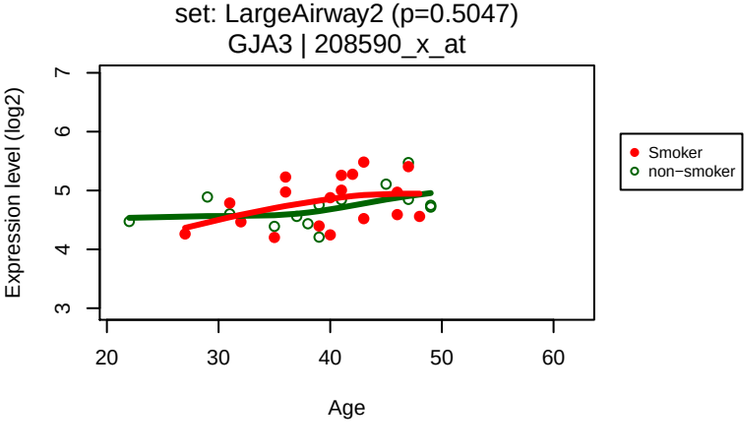

Gene expression ratio (smokers/nonsmokers)
gene: GJA3 | 208590_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2699
p=0.3089
p=0.8947
p=0.9425
p=0.5047
p=0.4175
p=0.415
p=0.7863
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

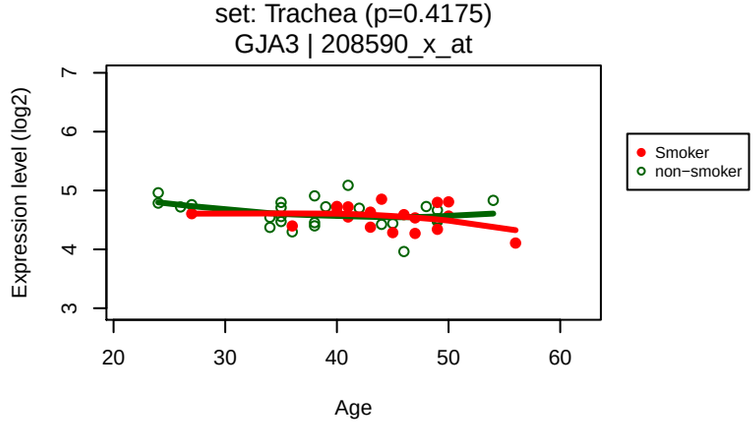

Gene expression ratio in Males
gene: GJA3 | 208590_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8258
p=0.3466
p=0.7633
p=0.4853
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GJA3 | 208590_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0741
p=0.7404
p=0.44
p=0.3016
SmallAirway2
LargeAirway2
Trachea
Macrophage2

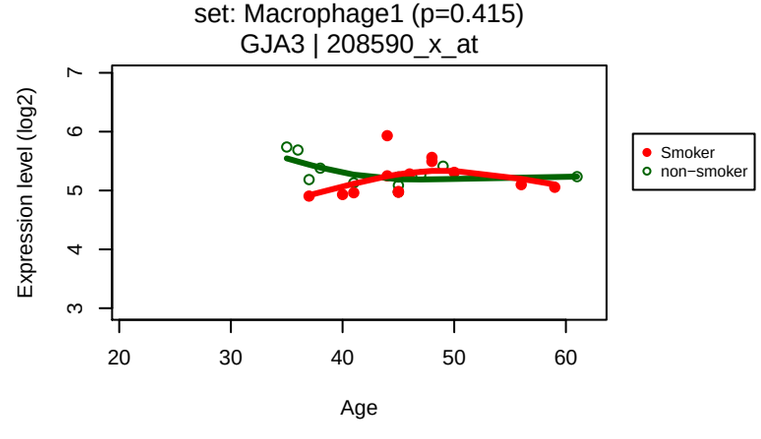

set: SmallAirway1 (p=0.2699)
GJA3 | 208590_x_at
Expression level (log2)
Age
Smoker
non−smoker

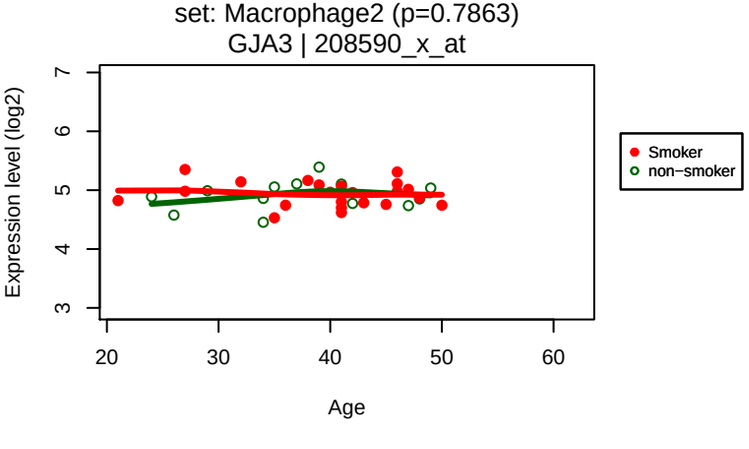

set: SmallAirway2 (p=0.3089)
GJA3 | 208590_x_at
Expression level (log2)
Age
Smoker
non−smoker

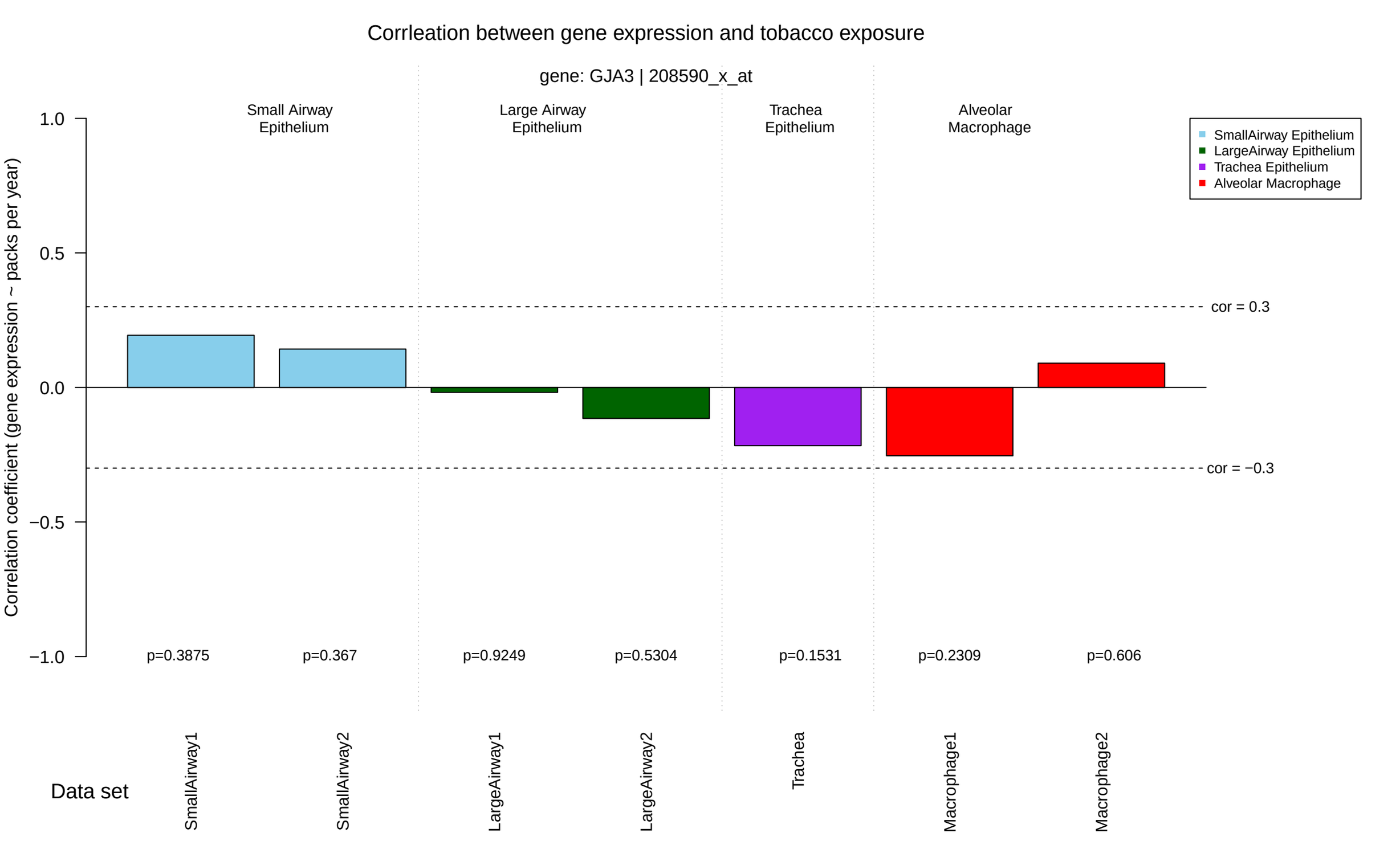

set: LargeAirway1 (p=0.9425)
GJA3 | 208590_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5047)
GJA3 | 208590_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4175)
GJA3 | 208590_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.415)
GJA3 | 208590_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7863)
GJA3 | 208590_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GJA3 | 208590_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3875
p=0.367
p=0.9249
p=0.5304
p=0.1531
p=0.2309
p=0.606
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2